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Unit 2 Artificial Intelligence in Our Lives

PDF Documents

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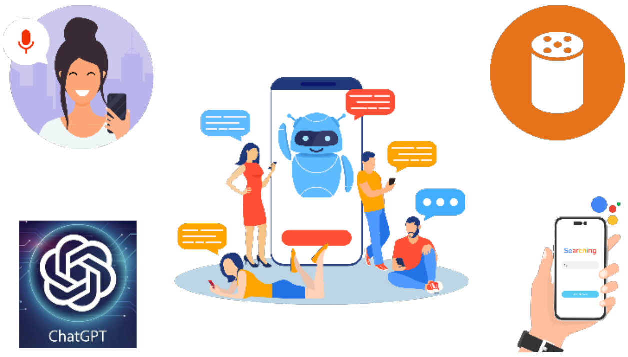
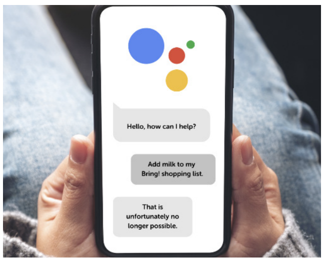
UNIT 2 ARTIFICIAL INTELLIGENCE IN OUR LIVES

INTRODUCING REAL-WORLD AI APPLICATIONS

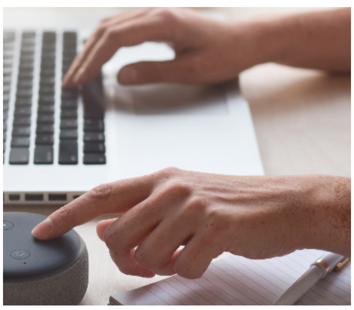
Artificial Intelligence (AI) is not just a concept from sci-fi movies; it's a reality that's part of our everyday lives. This unit explores how AI impacts various aspects of our daily routines, from how we communicate and learn to how we entertain ourselves and make decisions.



VIRTUAL ASSISTANTS: BRIDGING COMMUNICATION



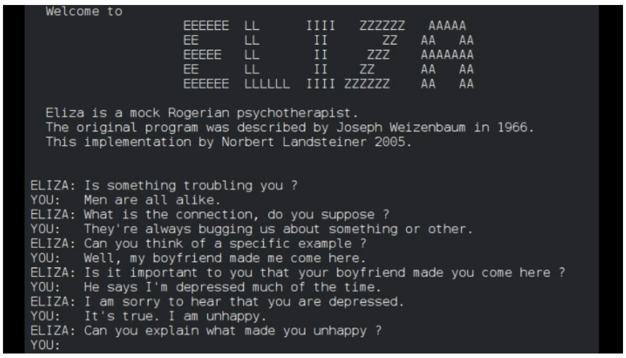
In the realm of AI, virtual assistants like Siri, Alexa, and Google Assistant represent a seamless blend of technology and daily convenience. They are engineered to understand spoken language, perform internet searches, manage schedules, and even control smart home devices. Through natural language processing, a branch of AI, these assistants decipher and execute verbal commands, making technology more accessible and intuitive.



THE EMERGENCE OF VIRTUAL ASSISTANTS

Virtual assistants, the conversational agents that assist us with day-to-day tasks, have roots in early AI research. The journey began in the 1960s with ELIZA, a simple program developed at MIT that could mimic human conversation by matching user input to pre-defined scripts. Although primitive, ELIZA laid the groundwork for natural language processing (NLP), a crucial AI domain.

The real leap towards modern virtual assistants came with advancements in NLP, machine learning, and the ubiquity of internet-connected devices. In 2011, Apple introduced Siri, the first widespread virtual assistant integrated into a smartphone, sparking a revolution in how we interact with our devices. Following Siri, major tech companies developed their own assistants, such as Amazon's Alexa (2014), Microsoft's Cortana (2014), and Google Assistant (2016).



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Welcome to
                  EEEEEE  LL      IIII  ZZZZZZ   AAAAA
                  EE      LL       II      ZZ   AA   AA
                  EEEEE   LL       II     ZZZ   AAAAAAA
                  EE      LL       II    ZZ     AA   AA
                  EEEEEE  LLLLLL  IIII  ZZZZZZ  AA   AA

  Eliza is a mock Rogerian psychotherapist.
  The original program was described by Joseph Weizenbaum in 1966.
  This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
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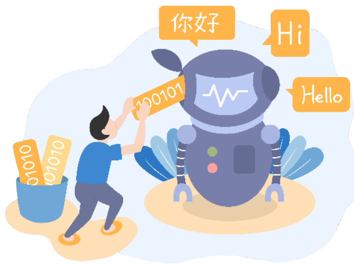
(ELIZA, developed from 1964 to 1967 at MIT
by Joseph Weizenbaum)

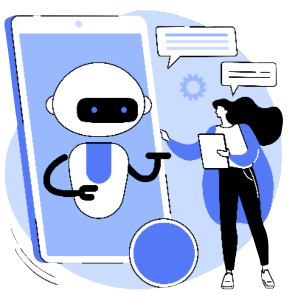
HOW VIRTUAL ASSISTANTS WORK



At their core, virtual assistants rely on NLP and voice recognition technology to understand spoken commands. When you ask a question or issue a command, the assistant converts your speech into text, processes the request using AI algorithms, and then performs the desired action, which could range from answering a question to controlling smart home devices.

The intelligence of these assistants comes from machine learning models trained on vast datasets, allowing them to understand a wide range of phrases and commands. Over time, they learn from interactions to improve their accuracy and expand their capabilities.

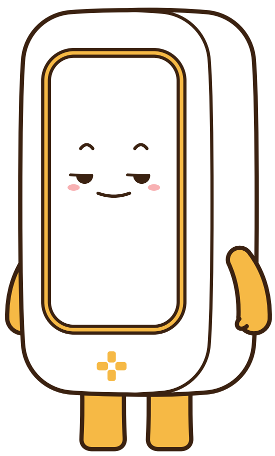
THE ROLE OF VIRTUAL ASSISTANTS TODAY



Virtual assistants have become integral to many aspects of modern life, from setting alarms and reminders to providing real-time information like weather updates and news. They also serve as interfaces for controlling smart home devices, making it easier to manage lighting, temperature, and security with simple voice commands.
In addition to personal use, virtual assistants are increasingly adopted in customer service and business environments, automating routine tasks and providing users with intuitive ways to interact with services and information.

FUTURE DIRECTIONS

The future of virtual assistants lies in becoming more proactive, personalized, and context-aware. The next generation of assistants aims to anticipate user needs, offer more personalized recommendations, and understand the context of requests better. Advances in AI, particularly in understanding human emotions and intentions, will enable virtual assistants to offer more nuanced and relevant responses, making them even more indispensable in our digital lives.



Discussion Questions

Discussion Question 1

Consider the impact of virtual assistants on daily life and human-computer interaction. In what ways have they changed how we access information and control devices?

Discussion Question 2

Discuss the potential future developments for virtual assistants. What are the ethical considerations and challenges that might arise as they become more integrated into our lives?